

Non-Graded Project

Module 1 – Cloud Foundations & Shared Responsibility

Building a Cloud Responsibility Matrix

Overview

Your organization is adopting a multi-cloud strategy. Leadership needs a clear, visual guide showing what the cloud provider secures vs. what the customer must secure.

Steps

- 1 **Choose a workload** (e.g., web app on EC2, database on RDS, SaaS app like Google Workspace).
- 2 **Map responsibilities** across IaaS, PaaS, and SaaS:
 - Networking, OS patching, IAM, data encryption, logging, etc.
- 3 **Design an artifact:** a responsibility matrix (table or infographic) showing who owns which security tasks.
- 4 **Peer Review:** exchange matrices with classmates—critique for completeness and clarity.

Outcome

A visual Shared Responsibility Matrix that could be shown to non-technical executives or used in onboarding new engineers.

Module 2 – Securing Compute, Network, and Storage

Designing a Secure Cloud Architecture

Overview

You've been tasked with proposing a secure environment for a new internal application. The solution must address AMI provenance, patching, firewalls, and data protection.

Steps

- 1 Draw a reference diagram including:**
 - Compute (EC2 with hardened AMIs)
 - Networking (restricted Security Groups, VPC subnets)
 - Storage (S3 with Block Public Access, RDS encrypted at rest)
- 2 Add security layers:** MFA, patch automation, IAM least privilege.
- 3 Document guardrails:** identify at least 3 Config or equivalent rules that enforce compliance.



Peer Review: have others suggest improvements or identify overlooked risks.

Outcome

An architecture diagram + policy document showing a layered defense-in-depth cloud design.

Module 3 – Detection, Incident Response, and Forensics

Cloud Incident Response Playbook

Overview

Your SOC just experienced a simulated IAM key leak. You must build a structured, repeatable **incident response (IR) playbook**.

Steps

- 1 **Scenario:** Developer posts AWS keys to GitHub. GuardDuty flags suspicious API calls.
- 2 **Write an IR playbook** covering:
 - Containment (disable key, isolate workloads)
 - Eradication & recovery (rotate credentials, rebuild from AMI)
 - Forensics (snapshotting, CloudTrail queries, evidence handling)
 - Lessons learned (post-mortem, control fixes)

- 3 **Include artifacts:** checklist, flow diagram, or runbook template.
- 4 **Peer Review:** classmates test your playbook against variations (e.g., EC2 compromise, S3 exfiltration).

Outcome

A step-by-step IR playbook demonstrating ability to operationalize detection and response.

Module 4 – Governance, Compliance, and Cost-Aware Security

Compliance & Cost Dashboard Prototype

Overview

Executives want proof that security controls both reduce risk and save money. You must design a dashboard that aligns CIS, ISO, or NIST safeguards with cost metrics.

Steps

- 1 **Pick 3–4 controls** (e.g., MFA enforcement, Block Public Access, Config conformance pack).
- 2 **Link controls to business outcomes:** e.g., “Enabling encryption satisfies CIS 3.1 and avoids regulatory penalties.”
- 3 **Design dashboard mockups:**
 - Security compliance status (per control)
 - Cost impact (e.g., right-sized instances, S3 lifecycle savings)
 - Tag-based accountability (Owner, DataClass).

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- 4 Peer Review:** classmates critique whether your dashboard “speaks to” both auditors and executives.

Outcome

A **dashboard concept (mockups, wireframe, or slides)** that ties compliance controls to **financial language** executives understand.

Instructions for Documenting and Sharing The Project for Peer Review

Document the Project



Use word processing software to create your report.

Ensure all sections of the project document structure are completed.

Proofread and edit your report for clarity and accuracy.



Share the Project

Document: Save the report in PDF format.

Upload: Share in the “Peer Review” area of the course shell with a brief description.



Peer Review

Review the projects submitted by your peers.

Provide constructive feedback and comments on their work.